

Revisiting Rational Broadcast Protocols

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Abstract. A recent study by Yamashita and Yasunaga (GameSec 2023) presented a constant-round deterministic broadcast protocol secure against *detection-averse* adversaries — those who prefer to attack without being detected. In this work, we revisit their protocol and observe that it remains secure even against a broader class of adversaries, not necessarily detection-averse. We formalize its detection mechanism as *local detectability* and construct broadcast protocols with local detectability that address two weaknesses of the original protocol: (1) it only guarantees weak validity, and (2) it may cause false detections. Our first protocol achieves round complexity four against rational adversaries and $t + 4$ against malicious adversaries, where the adversary corrupts at most t parties. Our second protocol achieves the optimal round complexity of $t + 1$ for malicious adversaries, while the round complexity is four against detection-averse adversaries.

1 Introduction

Byzantine broadcast is one of the most fundamental protocols in distributed computing and is used as a building block for various cryptographic protocols. In a broadcast protocol, a party called the *sender* wants to send its value m to all other parties. There are two requirements: (a) all honest parties output the same value (*agreement*), and (b) if the sender is honest, all honest (non-corrupted) parties output the sender’s value (*validity*). The *round complexity* is one of the most significant measures of the efficiency of protocols. We focus on the synchronous communication model in which the public-key infrastructure (PKI) is available.

In the seminal work by Dolev and Strong [5], they gave a $(t + 1)$ -round deterministic broadcast protocol against malicious adversaries who corrupt t out of n parties for any $t < n$. They also showed that the round complexity $t + 1$ is *optimal* among deterministic protocols. Since then, researchers have been focused on constructing *randomized* protocols for circumventing the lower bound of [5]. In the honest-majority setting ($t < n/2$), expected constant-round protocols have been proposed [6][12][1]. In the dishonest-majority setting ($t \geq n/2$), expected constant-round protocols have been presented for $t = n/2 + O(1)$ [9] and for $t = (1 - \epsilon)n$ [16] with $\epsilon \in (0, 1)$. However, it is known that constant-round randomized protocols cannot be constructed against malicious adversaries for $t = (1 - o(1))n$ [9].

A recent study by Yamashita and Yasunaga [17] explored a *game-theoretic* approach to broadcast protocols. Specifically, they studied a class of *rational* adversaries who aim to violate security requirements while avoiding detection. These adversaries, which we refer to as *detection-averse* adversaries, attack the protocol while considering whether their actions can be detected. (In [17], these adversaries are termed *timid*, but we adopt the term *detection-averse* as it more accurately captures their behavior.) The authors of [17] proposed a constant-round deterministic broadcast protocol secure against detection-averse adversaries for any $t < n$. Their protocol circumvents a round-complexity lower bound of $t + 1$ [5] by considering rational adversaries, a weaker adversary model than traditional Byzantine/malicious models.

In this work, we revisit the problem of constructing broadcast protocols against rational adversaries by analyzing the protocol of [17] (referred to as the YY23 protocol) in more depth. We begin by observing that the YY23 protocol is secure against rational adversaries *without relying on any attack detection mechanism*. That is, their security proofs go through even when the protocol has no detection mechanism. We can show that their protocol is secure against a broader class of rational adversaries, who have an incentive to attack the protocol but do not necessarily have the detection-averse property. For instance, this broader class includes adversaries who can still obtain utility even if detected — for example, when detection does not lead to immediate exclusion from the protocol, or when anonymity prevents any effective punishment. Such adversaries are rational but not detection-averse, and we observe that the protocol remains secure against them as well.

The mechanism for securing the YY23 protocol. We briefly explain why the YY23 protocol remains secure even against more general rational adversaries. The protocol has two possible termination points: round 4 and round $t + 4$.¹ It guarantees (weak) validity and agreement against malicious adversaries, in which case it terminates in round $t + 4$. Against rational adversaries, the protocol terminates early in round 4. This is because a rational adversary cannot improve its utility by violating the security requirements and has no incentive to *delay protocol termination*. Thus, its best response is to behave honestly, which results in early termination in round 4. In short, the early termination follows from the inherent security of the YY23 protocol rather than from its attack-detection mechanism.

Weaknesses of the YY23 protocol. Besides the relaxed adversary model, there are two weaknesses in the protocol of [17] when $t \geq n/2$. First, the YY23 protocol only achieves *weak validity* because if the sender is honest, all honest parties output the sender's value *or* $\perp$, the failure symbol. Moreover, an adversary

¹ The authors of [17] claimed that the round complexity of their protocol is 5 for rational adversaries and is $t + 5$ for malicious adversaries. However, we noticed that termination in round 5 can be incorporated into round 4. Thus, the actual termination rounds should be 4 and $t + 4$. We present a refined version of the YY23 protocol in Fig. 2.

Table 1. Comparison of Our Broadcast Protocols with the YY23 Protocol

Ref.	Resiliency	Local Detectability	Round Comp.	Adversary Model	Security Requirements
Π_{YY23} [17]	$t < n/2$	Yes	4	Rational	Agreement & Validity
	$t \geq n/2$	No	4	Rational	Agreement & Weak Validity
	$t < n$	No	$t + 4$	Malicious	Agreement & Weak Validity
Π_{LD}	$t < n$	Yes	$t + 4$	Rational Malicious	Agreement & Validity
Π_{DA}	$t < n$	Yes	$t + 1$	Detection Averse Malicious	Agreement & Validity Weak Agreement & Validity

can easily cause this weaker validity condition: if $t \geq n/2$ corrupted parties do nothing in the protocol, all honest parties will output $\perp$ even if the sender is honest. Second, the protocol may cause *false detections*. Namely, an honest party can be falsely detected as a cheater. Since the sender is the only party that can be detected as a cheater in the YY23 protocol, the adversary’s strategy of doing nothing in the protocol for $t \geq n/2$ results in the situation that the honest sender is declared a cheater. These two weaknesses may limit the use of the YY23 protocol as a building block of cryptographic protocols. In addition, the work of [17] did not formalize the property of attack detection. Thus, it is unclear what type of detection mechanism is employed in the YY23 protocol.

1.1 This Work

In this work, we have three main contributions.

First, we formalize a detection property of the YY23 protocol for minority corruption ($t < n/2$). We call this property *local detectability*: whenever any honest party i outputs a candidate list C_i of cheaters, every party named in C_i is in fact corrupted. Equivalently, no honest party that follows the protocol is never identified as a cheater by an honest party.

The term *local* implies that each candidate list C_i is retained by party i . Thus, this detection mechanism does not guarantee global agreement on the cheater list among honest parties. Nevertheless, local detectability can still be useful in practice. For example, if multiple parties from the same organization participate in the protocol, their local candidate lists $\{C_i\}_i$ can be shared within the group, serving as a deterrent against misbehavior. An advantage of local detectability over *global* detectability is its feasibility: as discussed in [13], identifying cheaters globally is impossible when the adversary controls a majority ($t \geq n/2$). In contrast, our protocol ensures local detectability even when the adversary corrupts a majority of the parties.

Second, we construct a constant-round broadcast protocol with local detectability that is secure against rational adversaries corrupting $t < n$ parties. The round complexity is four against rational adversaries and is $t + 4$ against malicious ones. In contrast to the YY23 protocol, our protocol satisfies *standard*

validity rather than *weak validity*. Moreover, since local detectability ensures no false detection, our protocol resolves both weaknesses of the YY23 protocol.

Third, we present a broadcast protocol with the optimal round complexity of $t + 1$ against malicious adversaries while maintaining round complexity four against *detection-averse* adversaries. That is, the detection-averse property enables this optimal round complexity. The protocol also satisfies local detectability. Since, as discussed earlier, the YY23 protocol does not require the detection-averse property, our protocol is the first to explicitly leverage this property in broadcast protocols.

In Table 1, we compare our protocols with the YY23 protocol Π_{YY23} in Fig. 2, where our first and second protocols are referred to as Π_{LD} and Π_{DA} , respectively.

1.2 Technical Overviews

In [17], the authors define security against rational adversaries as follows. For an n -party broadcast protocol Π and adversary A , a broadcast game is defined such that Π is executed where A is a player controlling t out of n parties. The outcome of the game consists of three binary values: $\text{validity}_{\text{fail}}$, $\text{agreement}_{\text{fail}}$, and undetected . The first two values represent whether A could violate validity and agreement of Π . The last one represents whether malicious actions of A were detected during the execution. They considered rational adversaries that have utilities such that the adversary prefers each of the three values to be satisfied. Then, Π is secure against rational adversaries if there is a harmless adversary B such that the expected utility of any rational adversary A is bounded above by that of B . This *best response* property ensures that every adversary has no incentive to deviate from the harmless strategy B .

Our observation on the YY23 protocol is that their security proof does not require the third value undetected . In other words, the YY23 protocol is secure against any rational adversary that prefers the two values $\text{validity}_{\text{fail}}$ and $\text{agreement}_{\text{fail}}$ to be satisfied. Note that this rational adversary is different from *malicious* adversaries, which include one that wants to delay the termination of the protocol. For such an adversary, the YY23 protocol terminates in round $t + 4$ and thus fails to achieve a constant-round execution.

We describe the construction ideas of our protocols. A key observation is that if the YY23 protocol is modified so that (standard) validity is satisfied for malicious adversaries, it satisfies local detectability, thus resolving false detection. Our goal is to modify the YY23 protocol to satisfy validity. To achieve this, we need a mechanism that allows the honest sender to send his message to all parties, even if $t \geq n/2$. Our idea is to perform the Dolev-Strong protocol [5] for sending the sender's message *in parallel with* the YY23 protocol. With this modification, each party may output a value in rounds 4, $t + 1$, or $t + 4$. One of the difficulties in designing our protocols is determining the termination rule. The termination rule of our first protocol Π_{LD} is that if each party fails to receive the message in round $t + 4$, the party outputs the value received by the Dolev-Strong protocol in round $t + 1$.

Our second protocol Π_{DA} achieves an optimal round complexity of $t + 1$ for malicious adversaries, while the round complexity for *detection-averse* adversaries is 4. The construction idea is that we halt the main (YY23) part in Π_{LD} after round 4. With this modification, the resulting protocol guarantees only *weak agreement*, which requires that every honest party outputs either some value m or $\perp$. This modified protocol is sufficient to achieve a constant-round execution for detection-averse adversaries. Since Π_{DA} satisfies validity for malicious adversaries, it achieves local detectability, just as Π_{LD} does.

1.3 Related Work

This work builds on the work by Yamashita and Yasunaga [17], which presented a constant-round protocol against detection-averse adversaries. As described above, Dolev and Strong [5] provided a deterministic broadcast protocol with optimal round complexity for malicious adversaries. Since then, many studies have been on constructing *randomized* protocols that aim to circumvent the round-complexity lower bound [5]. In cryptography, relaxed notions of broadcast protocols have been proposed. *Detectable broadcast* [7] has a mechanism for detecting malicious behavior, but, to the best of our knowledge, no constant-round protocol achieving this notion has been proposed. The notion of *broadcast with abort* was introduced by Goldwasser and Lindell [10], where the security requirements are weak agreement and weak validity. They showed that this weaker security notion can be used to construct secure multiparty computation. The notion of *covert security*, introduced in [3], guarantees that any deviation from the protocol can be detected with a constant probability. Thus, covert security may be sufficient to protect against detection-averse adversaries. However, to our knowledge, no prior work has focused on constructing broadcast protocols with covert security. The work of [3] gave secure multiparty computation protocols with covert security by assuming the existence of broadcast channels. As discussed in [15], covert security is not necessarily weaker than malicious security. In contrast, our security against rational adversaries is weaker than the standard malicious security (see Section 2.1 for discussion). *Early-stopping* broadcast protocols [4, 2, 14] guarantee that the protocol terminates earlier when the number of corrupted parties is less than the assumed upper bound t . The actual round complexity depends on the number of faults that occur during execution rather than on t . Although the existence of multiple termination points is similar to rational broadcast protocols studied in [17] and this work, our protocols ensure that early termination results from the strategic behavior of rational adversaries.

A game-theoretic notion of detection-averse (timid) adversaries was introduced in [8] for constructing efficient secure-message transmission (SMT) protocols. The work of [17] applied this notion to the problem of constructing broadcast protocols.

1.4 Organization

Section 2 introduces the network model, setup assumptions, and formal definitions of broadcast protocols. Sections 2.2 and 3 provide overviews of the Dolev-Strong protocol [5] and the YY23 protocol [17], respectively. Our first and second protocols are presented in Sections 4 and 5, respectively.

2 Preliminaries

We briefly describe the network model, setup assumptions, and the definition of Byzantine broadcast.

There are n parties in the network. We assume a synchronous communication model: the protocol proceeds in rounds, and in each round, every party can send messages to all other parties. Messages from non-corrupted (honest) parties are guaranteed to be delivered correctly before the next round begins.

We assume an authenticated setting where a public key infrastructure (PKI) and digital signature schemes are available. Each party can generate a signature using its secret key, and the validity of a signature can be verified using the corresponding public key.

A signature scheme consists of three algorithms ($\text{Gen}, \text{Sign}, \text{Ver}$). The key generation algorithm Gen takes a security parameter n as input and outputs a pair of keys (pk, sk) . The security parameter is usually represented by the string $1 \cdots 1$ of length n , denoted by 1^n . The signing algorithm Sign takes a secret key sk and a message m as input and outputs a signature σ . The verification algorithm Ver takes a public key pk and a pair (m, σ) as input and checks whether σ is a valid signature of m . As a correctness requirement, we require that $\text{Ver}(pk, m, \text{Sign}(sk, m)) = 1$ for any (pk, sk) generated by $\text{Gen}(1^n)$ and message m . For simplicity, we assume an ideal signature scheme for which no one can generate a pair (m, σ) satisfying $\text{Ver}(pk, m, \sigma) = 1$ without using sk , where the pair (pk, sk) is generated by $\text{Gen}(1^n)$.

In the authenticated setting, for a set of parties $[n] = \{1, \dots, n\}$, we assume that, for each $i \in [n]$, a key pair (pk_i, sk_i) is generated by $\text{Gen}(1^n)$, the secret key sk_i is given to i , and the set $\{pk_i\}_{i \in [n]}$ of public keys is available to all parties.

We introduce a *broadcast game* to define the requirements of broadcast protocols. It is a single-player game played by an adversary.

Definition 1 (Broadcast Game). *Given an n -party protocol Π and adversary A , the broadcast game $\mathsf{G}_{\Pi, A}^{\text{BC}}(n, t)$ of Π against A is defined as follows. Set the parameters $\text{validity}_{\text{fail}} = \text{agreement}_{\text{fail}} = 0$. The adversary A chooses a sender $s \in [n]$, a message m , and a set of corrupted parties $C \subseteq [n]$ satisfying $|C| \leq t$. Then, protocol Π is executed with $s \in [n]$ as a sender and m as the initial message of s . During the execution, the parties in C are controlled by A , while the others honestly follow the description of Π . After protocol execution, each party $i \in [n]$ outputs val_i , the output value of party i . Let $H = [n] \setminus C$.*

- If $s \in H$ and there exists $i \in H$ such that $\text{val}_i \neq m$, set $\text{validity}_{\text{fail}} = 1$.

- If there exist different $i, j \in H$ such that $\text{val}_i \neq \text{val}_j$, set $\text{agreement}_{\text{fail}} = 1$.

The outcome of the game is $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$.

The traditional Byzantine broadcast is defined as follows. We refer to an adversary that controls at most t parties as a t -adversary.

Definition 2 (Byzantine Broadcast). An n -party protocol Π is said to satisfy *validity* (respectively, *agreement*) against t -adversaries if for every t -adversary A , the outcome $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$ of the broadcast game $\mathbf{G}_{\Pi, A}^{\text{BC}}(n, t)$ satisfies $\text{validity}_{\text{fail}} = 0$ (respectively, $\text{agreement}_{\text{fail}} = 0$). We say Π is a t -resilient broadcast protocol if Π satisfies *validity* and *agreement* against t -adversaries.

2.1 Rational Broadcast Protocols

In this section, we present our framework of the rational adversary model, which mainly follows that in [17]. We use the broadcast game of Definition 1 as a single-player game played by a rational adversary. Note that the information on whether the attack was detected, which was adopted in [17], is not included in the outcome of Definition 1.

We define the utility function of rational adversaries as follows.

Definition 3 (Utility Function). A utility function is a function that maps outcome $\text{out} = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$ of a broadcast game to a real value. A rational t -adversary has a utility function U satisfying

1. $U(\text{out}) > U(\text{out}')$ if $\text{validity}_{\text{fail}} > \text{validity}'_{\text{fail}}$ and $\text{agreement}_{\text{fail}} = \text{agreement}'_{\text{fail}}$;
2. $U(\text{out}) > U(\text{out}')$ if $\text{validity}_{\text{fail}} = \text{validity}'_{\text{fail}}$ and $\text{agreement}_{\text{fail}} > \text{agreement}'_{\text{fail}}$,

where $\text{out} = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$ and $\text{out}' = (\text{validity}'_{\text{fail}}, \text{agreement}'_{\text{fail}})$ are outcomes of the broadcast game.

Namely, rational adversaries gain higher utility if they violate the security requirements of broadcast protocols. The following relation holds:

$$U(0, 0) < \min\{U(0, 1), U(1, 0)\} \leq \max\{U(0, 1), U(1, 0)\} < U(1, 1). \quad (1)$$

The adversary's utility when following strategy A is the expected value of the utility when the adversary plays the broadcast game by following strategy A .

Definition 4 (Expected Utility). The utility $u(A)$ of t -adversary A is the expected value $\mathbb{E}[U(\text{out}_A)]$, where the expectation is taken over the randomness in the broadcast game $\mathbf{G}_{\Pi, A}^{\text{BC}}(n, t)$, including random coins of adversary A .

We define the security notion of broadcast protocols against a rational adversary based on the adversary's utility function. This formulation was also employed in [11, 8] for other protocols.

Definition 5 (Security of Rational Broadcast). *A broadcast protocol Π is said to be secure against rational t -adversaries if there exists a t -adversary B such that*

1. *Security: The outcome of $\mathbf{G}_{\Pi,B}^{\text{BC}}(n, t)$ satisfies $\text{out}_B = (0, 0)$.*
2. *Best Response: For any t -adversary A , the utility satisfies $u(A) \leq u(B)$.*

The round complexity against rational adversaries is the number of rounds required to terminate the execution of Π in $\mathbf{G}_{\Pi,B}^{\text{BC}}(n, t)$. The round complexity against worst-case adversaries is the number of rounds required to terminate the execution of Π in $\mathbf{G}_{\Pi,A}^{\text{BC}}(n, t)$ for any t -adversary A .

The first condition implies that adversary B does not violate the protocol's security requirements. The second ensures that B 's strategy is optimal among all possible strategies of t -adversaries, meaning that no adversary A can gain a higher expected utility. Thus, if Π satisfies the two conditions, all rational adversaries have no incentive to deviate from the harmless strategy B .

We remark that the security against rational adversaries (Definition 5) is weaker than the traditional security against malicious adversaries (Definition 2). Suppose Π is a t -resilient broadcast protocol. Then, every t -adversary A satisfies $u(A) = U(0, 0)$. Thus, by choosing any B in Definition 5, we can show that Π is secure against rational t -adversaries.

2.2 Dolev-Strong Protocol

Let us recall the Dolev-Strong authenticated broadcast protocol [5]. The protocol uses a *signature chain*. A signature chain for value v of length ℓ is defined as (1) $(v, \sigma_i(v))$ for some $i \in [n]$ if $\ell = 1$; (2) $(c, \sigma_i(c))$ for some $i \in [n]$ for $\ell > 1$, where c is a signature chain for v of length $\ell - 1$ that consists of signatures with $\ell - 1$ distinct signers other than i . A signature chain is valid if it satisfies the above conditions and all signatures are valid.

In the first round of the Dolev-Strong protocol, the sender with message m sends $(m, \sigma_s(m))$ to all parties. In round $r \geq 2$, each party i sends a signature chain of length r to all parties by appending his signature to a chain of length $r - 1$ received in the previous round. After receiving chains in round $t + 1$, if party i knows valid chains only for message m , i outputs m . Otherwise, i outputs $\perp$. This protocol has the property that once an honest party receives a valid signature chain for message m , every honest party will receive a valid chain for m in the later rounds. Hence, validity immediately follows. Since every valid signature chain sent in round r contains signatures of r different parties and the protocol continues until round $t + 1$, there exists an honest party who sent a valid chain in round $r \in [t + 1]$ (Otherwise, every honest party receives no valid chain and outputs $\perp$). Thus, after round $t + 1$, every honest party receives valid chains for the same messages, implying agreement.

The protocol description is presented in Fig. 1. Protocol $\Pi_{\text{DS}}(n, t, s)$ is a t -resilient broadcast protocol (Definition 2) for any $t < n$ [5].

Protocol $\Pi_{DS}(n, t, s)$

This protocol is parameterized by the number of parties n , the threshold t with $t < n$, and the sender index $s \in [n]$. The protocol assumes a PKI, where each $i \in [n]$ can generate a signature $\sigma_i(m)$ for message m by $\text{Sign}(sk_i, m)$ and can verify the validity of a signature σ of party j for message m' by $\text{Ver}(pk_j, m', \sigma)$.

- Round 1: Each party $i \in [n]$ sets $V_i = \emptyset$. Sender s with input m sends $(m, \sigma_s(m))$ to all parties.
- Round $r = 2, \dots, t + 1$: Each party $i \in [n]$ do the following:
 - If party i receives a valid signature chain c for value v of length $r - 1$ that does not contain i 's signature, i adds v to V_i and sends $(c, \sigma_i(c))$ to all parties. (Party i performs this procedure once for each value v . Thus, once v is added to V_i , i does nothing for v thereafter.)
 - **Termination:** At the end of round $t + 1$, if $|V_i| = 1$, i outputs $v \in V_i$. Otherwise, i outputs $\perp$.

Fig. 1. Dolev-Strong Protocol

3 YY23 Protocol

In this section, we review the YY23 protocol [17], which serves as the basis of our protocols. It satisfies agreement in Definition 2 and *weak validity*, meaning that if the sender is honest, all honest parties output the sender's value or $\perp$. The protocol uses several notions listed below:

- A *countersignature* is a signature for a signature: $(m, \sigma_i(m), \sigma_j(\sigma_i(m)))$, which is a signature chain of length 2 in the Dolev-Strong protocol, except that i and j are not necessarily distinct. A countersignature

$$\text{CSig}_m^i = (m, \sigma_s(m), \sigma_i(\sigma_s(m))),$$

can serve as a certificate that the sender s sent message m to party i .

- A *Proof of Dissemination (PoD)* is a signed set of $t + 1$ valid countersignatures:

$$\text{PoD}_m^i = (\text{PoD}_m = \{\text{CSig}_m^j\}_j, \sigma_i(\text{PoD}_m)).$$

In the protocol, it serves as a certificate that party i knows that every party obtained a countersignature for message m .

- A *Proof of Agreement (PoA)* is a signed set of $t + 1$ valid PoDs:

$$\text{PoA}_m^i = (\text{PoA}_m = \{\text{PoD}_m^j\}_j, \sigma_i(\text{PoA}_m)).$$

It serves as a certificate that party i knows that every party knows that every party obtained a countersignature for message m .

In the first round of the YY23 protocol, the sender s sends a signed message $(m, \sigma_s(m))$ to all parties. In round 2, each party sends a countersignature to

all parties. In round 3, each party generates a PoD from countersignatures sent in the previous round and sends a PoD to all parties. In round 4, each party generates a PoA from PoDs and sends a PoA to all parties via the Dolev-Strong protocol. At the end of round 4, if each party collected $t + 1$ valid PoAs for message m , it outputs m and halts. Otherwise, the party continues the DS protocol and outputs the value sent through the DS protocol after round $t + 4$. If party i fails to obtain a PoA, i outputs $\perp$ and considers the sender s as cheating.

We briefly describe a reason why this protocol works. We can show the following two claims:

- (1) PoAs can be generated only for a single message.
- (2) If some honest party outputs $\perp$, then all honest parties output $\perp$.

To prove (1), suppose PoAs exist for different m and m' . Since a PoA contains $t + 1$ PoDs, at least one PoD was sent by an honest party, who sent the PoD to all parties in round 3. Thus, all honest parties should have received PoDs for both m and m' . Since each valid PoD contains $t + 1$ countersignatures, all honest parties should have received countersignatures for both m and m' . However, in that case, honest parties send nothing in round 3, a contradiction. (Note that PoDs can be generated for both m and m' .) To prove (2), note that an honest party outputs $\perp$ only when that party failed to receive a PoA via the DS protocol. Since the DS protocol satisfies validity, it implies that no honest party sent a PoA using the DS protocol. If some honest party received a PoA, then all honest parties receive the same PoA by the agreement property of the DS protocol. Thus, (2) follows.

To prove weak validity, suppose that an honest party outputs a message m' different from the original message m of the honest sender. However, this is impossible because the sender's signature cannot be generated by any party other than the sender. To prove agreement, suppose that two honest parties output different messages m and m' . If neither m nor m' is $\perp$, it implies that PoAs for different messages were generated, contradicting (1). If $m' = \perp$, (2) implies that a PoA for message m was never sent to honest parties, a contradiction.

We provide a protocol description of the YY23 protocol in Fig. 2.

By the same argument as in [17], we can show that $\Pi_{YY23}(n, t, s)$ is secure against rational t -adversaries, where the security requirements of broadcast protocols are agreement and *weak validity* in the sense that an honest party may output $\perp$ even when a sender is honest. This security notion is formalized by modifying Definition 5 such that (a) Π satisfies weak validity and (standard) agreement against B , and (b) in the broadcast game of Definition 1, val_{fail} is set to be 1 when there exists $i \in H$ such that $\text{val}_i \notin \{m, \perp\}$. If protocol Π satisfies the above, we say Π is *weak-validity secure* against rational t -adversaries.

Theorem 1. *For any $t < n$, protocol Π_{YY23} is weak-validity secure against rational t -adversaries. The round complexity is 4 against rational t -adversaries and is $t + 4$ against worst-case t -adversaries.*

The proof of Theorem 1 is almost the same as that of [17, Theorem 1].

Protocol $\Pi_{YY23}(n, t, s)$

This protocol is parameterized by the number of parties n , the threshold t with $t < n$, and the sender index $s \in [n]$. The protocol assumes a PKI, where each $i \in [n]$ can generate a signature $\sigma_i(m)$ for message m by $\text{Sign}(sk_i, m)$ and can verify the validity of a signature σ of party j for message m' by $\text{Ver}(pk_j, m', \sigma)$. We denote by $\Pi_{DS}^r(n, t, s)$ the protocol description of the r th round of $\Pi_{DS}(n, t, s)$. The validity conditions of signature, countersignature, PoD, and PoA are summarized in Table 2.

- Round 1: Sender s with input m sends $(m, \sigma_s(m))$ to all parties.
- Round 2: Each party $i \in [n]$ checks whether it has received a valid signature $(m, \sigma_s(m))$ from s and no valid signature for any other value $m' \neq m$ in round 1. If so, i sends

$$\text{CSig}_m^i = (m, \sigma_s(m), \sigma_i(\sigma_s(m)))$$

to all parties. Otherwise, i sends nothing.

- Round 3: Each party $i \in [n]$ checks whether it has received $t + 1$ valid countersignatures CSig_m^j for the same value m , each from a distinct signer j , and no valid countersignature for any other value $m' \neq m$ in round 2. If so, i sends

$$\text{PoD}_m^i = \left(\text{PoD}_m = \{ \text{CSig}_m^j \}_j, \sigma_i(\text{PoD}_m) \right)$$

to all parties, where PoD_m consists of $t + 1$ valid countersignatures. Otherwise, i sends nothing.

- Round 4: Each party $i \in [n]$ checks whether it has received $t + 1$ valid PoD_m^j for the same value m , each from a distinct signer j , in round 3. If so, i prepares

$$\text{PoA}_m^i = \left(\text{PoA}_m = \{ \text{PoD}_m^j \}_j, \sigma_i(\text{PoA}_m) \right),$$

where PoA_m consists of $t + 1$ valid PoDs, and performs $\Pi_{DS}^1(n, t, i)$ where the input message of i is PoA_m^i . Otherwise, i sends nothing.

Termination: At the end of round 4, each party $i \in [n]$ checks whether it has received $t + 1$ valid PoA_m^j for the same message, each from a distinct signer j . If so, i outputs m and halts. Otherwise, i continues with the later rounds.

- Round $r = 5, \dots, t + 4$: Each party $i \in [n]$ performs $\Pi_{DS}^{r-3}(n, t, j)$ for every $j \in [n]$.

Termination: At the end of round $t + 4$, for each $i \in [n]$, let V_i be the set of values output by party i via $\Pi_{DS}(n, t, j)$ for $j \in [n]$. If V_i contains valid PoA_m^j for a single value m , then i outputs m . Otherwise, i outputs $\perp$.

Cheater Detection: For each $i \in [n]$, if party i outputs $\perp$ in termination, then i outputs $C_i = \{s\}$ as a candidate list of cheaters.

Fig. 2. YY23 Protocol

Table 2. Validity Conditions

Objects	Validity Conditions
Signature $\sigma_i(m)$	$\text{Ver}(pk_i, m, \sigma_i(m)) = 1$
Countersignature $(m, \sigma_s(m), \sigma_i(\sigma_s(m)))$	$\sigma_s(m)$ and $\sigma_i(\sigma_s(m))$ are valid signatures
Proof of dissemination (PoD) $\text{PoD}_m^i = (\text{PoD}_m = \{\text{CSig}_m^j\}_j, \sigma_i(\text{PoD}_m))$	$\sigma_i(\text{PoD}_m)$ is a valid signature $\wedge$ every CSig_m^j is a valid countersignature $\wedge$ each j of CSig_m^j is distinct $\wedge \text{PoD}_m = t + 1$
Proof of agreement (PoA) $\text{PoA}_m^i = (\text{PoA}_m = \{\text{PoD}_m^j\}_j, \sigma_i(\text{PoA}_m))$	$\sigma_i(\text{PoA}_m)$ is a valid signature $\wedge$ every PoD_m^j is a valid PoD $\wedge$ each j of PoD_m^j is distinct $\wedge \text{PoA}_m = t + 1$

4 Protocols with Local Detectability

In this section, we define the property of local cheating detectability. To this end, we consider an extension of the broadcast game of Definition 1, where each party $i \in [n]$ can output a candidate list $C_i \subseteq [n]$, indicating the parties that party i suspects of cheating. We denote the game by $G_{\Pi, A}^{\text{BC-LD}}(n, t)$.

Definition 6 (Local Detectability). *An n -party broadcast protocol Π is said to have local detectability against t -adversaries if for every t -adversary A and every execution of the broadcast game $G_{\Pi, A}^{\text{BC-LD}}(n, t)$, the following holds:*

- If $i \in H$ outputs a candidate list C_i of cheaters, then $C_i \subseteq C$.

Here, $C \subseteq [n]$ is a set of corrupted parties in the broadcast game and $H = [n] \setminus C$.

Note that the above is equivalent to the following condition:

- If $i \in H$, then no party $j \in H$ outputs C_j such that $i \in C_j$.

As local detectability, we require only *detection soundness* (no false positives). We do not require *detection completeness* (no false negatives), since it is impossible to detect corrupted parties that follow the protocol honestly.

As discussed in [17], Π_{YY23} satisfies local detectability for $t < (n - 1)/2$, whereas the condition fails for $t \geq (n - 1)/2$.

In this section, we present a protocol that achieves local detectability for any $t < n$. As outlined in Section 1, the idea is to perform the Dolev-Strong protocol and the YY23 protocol in parallel. One difficulty in designing such protocols is defining an appropriate termination rule. The description of our protocol Π_{LD} is given in Fig. 3.

4.1 Achieving Local Detectability

We present sufficient conditions for achieving local detectability.

Protocol $\Pi_{LD}(n, t, s)$

This protocol is parameterized by the number of parties n , the threshold t with $t < n$, and the sender index $s \in [n]$. The protocol assumes a PKI as in $\Pi_{YV23}(n, t, s)$. We denote by $\Pi_{DS}^r(n, t, s)$ the protocol description of the r th round of $\Pi_{DS}(n, t, s)$. We use $\overline{\Pi_{DS}^r}(n, t, s)$ as the description of the r th round of $\Pi_{DS}(n, t, s)$ for performing the auxiliary part. The validity conditions are summarized in Table 2.

- Round 1: Sender s with input m sends $(m, \sigma_s(m))$ to all parties.
- Round 2: Henceforth, each party runs the auxiliary part (denoted by [Aux]) in parallel with the main part, described without an explicit label. Each party $i \in [n]$ checks whether it has received a valid signature $(m, \sigma_s(m))$ from s and no valid signature for any other value $m' \neq m$ in round 1. If so, i sends

$$\text{CSig}_m^i = (m, \sigma_s(m), \sigma_i(\sigma_s(m)))$$

to all parties. Otherwise, i sends nothing.

[Aux] Each party $i \in [n]$ regards the messages received in round 1 as those of $\Pi_{DS}^1(n, t, s)$ and then performs $\overline{\Pi_{DS}^2}(n, t, s)$.

- Round 3: Each party $i \in [n]$ checks whether it has received $t + 1$ valid countersignatures CSig_m^j for the same value m , each from a distinct signer j , and no valid countersignature for any other value $m' \neq m$ in round 2. If so, i sends

$$\text{PoD}_m^i = \left(\text{PoD}_m = \{ \text{CSig}_m^j \}_j, \sigma_i(\text{PoD}_m) \right)$$

to all parties. Otherwise, i sends nothing.

[Aux] If $t \geq 2$, each party $i \in [n]$ performs $\overline{\Pi_{DS}^3}(n, t, s)$.

- Round 4: Each party $i \in [n]$ checks whether it has received $t + 1$ valid PoD_m^j for the same value m , each from a distinct signer j , in round 3. If so, i prepares

$$\text{PoA}_m^i = \left(\text{PoA}_m = \{ \text{PoD}_m^j \}_j, \sigma_i(\text{PoA}_m) \right).$$

and performs $\Pi_{DS}^1(n, t, i)$ where the input message of i is PoA_m^i . Otherwise, i sends nothing.

Termination: At the end of round 4, each party $i \in [n]$ checks whether it has received $t + 1$ valid PoA_m^j for the same value m , each from a distinct signer j . If so, i outputs m and halts both the main and the auxiliary parts. Otherwise, i continues with the auxiliary part.

[Aux] If $t \geq 3$, each party $i \in [n]$ performs $\overline{\Pi_{DS}^4}(n, t, s)$.

- Round $r = 5, \dots, t + 4$: Each $i \in [n]$ performs $\Pi_{DS}^{r-3}(n, t, j)$ for every $j \in [n]$.

[Aux] In round $r \leq t + 1$, each party $i \in [n]$ performs $\overline{\Pi_{DS}^r}(n, t, s)$. Let v_{aux}^i be the output of party i by performing the auxiliary part $\overline{\Pi_{DS}^r}(n, t, s)$.

Termination: At the end of round $t + 4$, for each $i \in [n]$, let V_i be the set of values output by party i via $\Pi_{DS}(n, t, j)$ for $j \in [n]$. If V_i contains valid PoA_m^j for a single value m , then i outputs m . Otherwise, i outputs v_{aux}^i .

Cheater Detection: For each $i \in [n]$, if party i outputs $\perp$ in termination, then i outputs $C_i = \{s\}$ as a candidate list of cheaters.

Fig. 3. Our Protocol with Local Detectability

Lemma 1. *Suppose that (a) a broadcast protocol Π satisfies validity against every t -adversary and that (b) every honest party i in Π outputs a candidate list C_i of cheaters with $|C_i| \geq 1$ only when i outputs $\perp$ and $C_i = \{s\}$. Then, Π achieves local detectability against t -adversaries.*

Proof. Suppose $i \in H$ outputs a candidate list C_i with $|C_i| \geq 1$. Assumption (b) implies that i outputs $\perp$ and $C_i = \{s\}$. If $s \in H$, assumption (a) implies that every $i \in H$ outputs the message m of s . That means $i \in H$ outputs m , a contradiction. Hence, $s \in C$. Therefore, $C_i \subseteq C$. $\square$

We show that protocol Π_{LD} satisfies the conditions in Lemma 1, implying that Π_{LD} has local detectability.

Theorem 2. *For any $t < n$, the broadcast protocol Π_{LD} has local detectability against t -adversaries.*

Proof. It is sufficient to show that Π_{LD} satisfies (a) and (b) in Lemma 1.

First, we prove (a), validity of Π_{LD} . We need to show that when s is honest and has input m , every honest party outputs m in the protocol. By the validity of the DS protocol, it must be the case that $v_{\text{aux}}^i = m$ for every honest party i . Then, if honest party i outputs $m' \neq m$, it must be either of the following two cases:

1. At the end of round 4, i collected $t + 1$ valid $\text{PoA}_{m'}^j$ for different j .
2. At the end of round $t + 4$, it holds that $V_i = \{\text{PoA}_{m'}^j\}$.

However, since s is honest, $\sigma_s(m')$ cannot be generated, and thus neither of the above two cases can occur. Hence, Π_{LD} satisfies validity.

Condition (b) immediately follows from the protocol description of Π_{LD} . $\square$

4.2 Proving Security against Rational Adversaries

We show that Π_{LD} is a secure broadcast protocol against rational adversaries.

The following is a useful lemma claiming that PoAs can be generated only for a single message in Π_{LD} .

Lemma 2. *In every broadcast game $G_{\Pi_{\text{LD}}, A}^{\text{BC-LD}}(n, t)$ of protocol Π_{LD} against t -adversary A , if any party holds a valid proof of agreement PoA_m^i for some m , then no party can obtain a valid proof of agreement $\text{PoA}_{m'}^j$ for any $m' \neq m$.*

Proof. Suppose some party obtained a valid PoA_m^i for m . Since it contains $t + 1$ valid PoDs for m and at most t parties are corrupted, at least one PoD_m^j was generated by honest party j . Since PoD_m^j contains $t + 1$ valid countersignatures for m , at least one countersignature CSig_m^k was generated by honest party k . Thus, all parties received this countersignature CSig_m^k sent from party k .

Suppose for contradiction that some party obtained a valid $\text{PoA}_{m'}^j$ for $m' \neq m$. By the same argument, all parties received a valid countersignature for m' . However, in that case, every honest party ℓ should have sent nothing in round

3 since party ℓ saw valid countersignatures for different messages m and m' . It contradicts the fact that valid PoDs for m and m' exist. Therefore, the statement follows. $\square$

We prove that Π_{LD} satisfies agreement.

Lemma 3. *For any $t < n$, Π_{LD} satisfies agreement for t -adversaries.*

Proof. We need to show that for any t -adversary A , the outcome $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$ of the broadcast game $G_{\Pi_{LD}, A}^{\text{BC}}(n, t)$ satisfies $\text{agreement}_{\text{fail}} = 0$.

Suppose for contradiction that there exists a t -adversary A such that $\text{agreement}_{\text{fail}} = 1$ in $G_{\Pi_{LD}, A}^{\text{BC}}(n, t)$. It implies that there exist different $i, j \in H$ such that $\text{val}_i \neq \text{val}_j$. There are two cases: $(\text{val}_i, \text{val}_j) = (m, \perp)$ and $(\text{val}_i, \text{val}_j) = (m, m')$ for $m \neq m'$, $\perp \notin \{m, m'\}$. We show that the two cases cannot occur.

We observe that $\text{val}_i = m$ occurs either in the following three cases:

- (a) Party i outputs m in round 4 by collecting $t + 1$ valid PoA_m^ℓ for different ℓ .
- (b) Party i outputs m in round $t + 4$ since V_i contains valid PoA_m^ℓ only for m .
- (c) Party i outputs m in round $t + 4$ since V_i contains no valid PoA and $v_{\text{aux}}^i = m$.

Note that $\text{val}_j = \perp$ occurs only in the following case:

- (d) Party j outputs $\perp$ in round $t + 4$ since V_j contains no valid PoA and $v_{\text{aux}}^j = \perp$.

First, consider case (a). Since i has $t + 1$ valid PoA_m^ℓ for different ℓ , one of them is a valid $\text{PoA}_m^{\ell'}$ for $\ell' \in H$. Since ℓ' had sent $\text{PoA}_m^{\ell'}$ to all parties in round 4, every honest party k added $\text{PoA}_m^{\ell'}$ to V_k via $\Pi_{DS}(n, t, \ell')$. Lemma 2 implies that no honest party could obtain valid $\text{PoA}_{m'}$ for $m' \neq m$. Thus, V_j contains valid PoAs only for m , implying that $m' = m$, a contradiction.

Next, consider case (b). Since i receives a valid PoA_m^ℓ via $\Pi_{DS}(n, t, \ell)$ for some ℓ , every honest party should receive PoA_m^ℓ by the agreement property of Π_{DS} . Since there is no PoAs for $m' \neq m$ by Lemma 2, V_j contains valid PoAs only for m , implying $m' = m$, a contradiction.

Finally, consider case (c). It is sufficient to consider the case that party j outputs m' in round $t + 4$, in which case V_j contains no valid PoA and $v_{\text{aux}}^j \in \{m', \perp\}$. However, since $i, j \in H$, the agreement property of $\Pi_{DS}(n, t, s)$ implies that $v_{\text{aux}}^j = m$, which contradicts $v_{\text{aux}}^j \in \{m', \perp\}$. $\square$

Next, we show that Π_{LD} satisfies validity.

Lemma 4. *For any $t < n$, Π_{LD} satisfies validity for t -adversaries.*

Proof. To prove that Π_{LD} satisfies validity, we need to show that for any t -adversary A , the outcome $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}})$ of the broadcast game $G_{\Pi_{LD}, A}^{\text{BC}}(n, t)$ satisfies $\text{validity}_{\text{fail}} = 0$. Namely, if sender s with initial message m is not corrupted, every $i \in H$ outputs $\text{val}_i = m$.

Suppose for contradiction that there exists $i \in H$ such that $\text{val}_i = m' \neq m$. Since $s \in H$, the validity of the DS protocol $\Pi_{DS}(n, t, s)$ in the auxiliary part

implies that $v_{\text{aux}}^j = \{m\}$ for every $j \in H$. By the protocol description of Π_{LD} , it must be the case that i receives a valid $\text{PoA}_{m'}^\ell$ in the execution. However, the existence of $\text{PoA}_{m'}^\ell$ requires a valid signature of s for message m' . That is impossible as long as the signature scheme is secure. Hence, the statement follows. $\square$

We show that Π_{LD} is a constant-round broadcast protocol against rational adversaries.

Theorem 3. *For any $t < n$, the broadcast protocol Π_{LD} is secure against rational t -adversaries. It has round complexity 4 against rational adversaries and $t + 4$ against worst-case adversaries.*

Proof. Consider a t -adversary B that chooses a random sender $s \in [n]$, a random message m , and $C = \emptyset$. In this case, B does not perform any attacks on the protocol. Since each party i receives a valid PoA_m^j from all n parties in round 4 and outputs m in the same round, the outcome of the broadcast game $G_{\Pi, B}^{\text{BC}}(n, t)$ is $\text{out}_B = (0, 0)$.

Next, we show the best-response property. Since $u(B) = U(0, 0)$, we need to show that $u(A) \leq U(0, 0)$ for any t -adversary A . Lemmas 3 and 4 imply that Π_{LD} achieves validity and agreement for any t -adversary. It implies that every t -adversary A achieves $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}}) = (0, 0)$, meaning that $u(A) = U(0, 0) \leq u(B)$.

The worst-case round complexity is $t + 4$ by the protocol description. $\square$

5 Protocols with Optimal Worst-Case Round Complexity

We show that if rational t -adversaries are *detection averse*, we can construct a broadcast protocol that achieves the optimal worst-case round complexity of $t + 1$.

Before presenting the protocol, we formally define the security notion. First, we extend the broadcast game $G_{\Pi, A}^{\text{BC-LD}}(n, t)$ introduced in Section 4 to include an additional parameter, **undetected**, which indicates whether the adversary's attack is locally detected by honest parties.

Definition 7 (Extended Broadcast Game with Local Detectability).

Given an n -party protocol Π and adversary A , the broadcast game $G_{\Pi, A}^{\text{eBC-LD}}(n, t)$ of Π against A is defined as follows. Set the parameters $\text{validity}_{\text{fail}} = \text{agreement}_{\text{fail}} = 0$ and $\text{undetected} = 1$. The adversary A chooses a sender $s \in [n]$, a message m , and a set of corrupted parties $C \subseteq [n]$ satisfying $|C| \leq t$. Then, protocol Π is executed with $s \in [n]$ as a sender and m as the initial message. During the execution, the parties in C are controlled by A , while the others honestly follow the description of Π . After protocol execution, each party $i \in [n]$ outputs a value val_i and a candidate list $C_i \subseteq [n]$. Let $H = [n] \setminus C$.

- If $s \in H$ and there exists $i \in H$ such that $\text{val}_i \neq m$, set $\text{validity}_{\text{fail}} = 1$.
- If there exist different $i, j \in H$ such that $\text{val}_i \neq \text{val}_j$, set $\text{agreement}_{\text{fail}} = 1$.

- If there exist $i \in C$ and $j \in H$ such that $i \in C_j$, set $\text{undetected} = 0$.

The outcome of the game is $\text{out}_A = (\text{validity}_{\text{fail}}, \text{agreement}_{\text{fail}}, \text{undetected})$.

The parameter undetected is initially set to 1, and is changed to 0 if there exists an honest party j that outputs a candidate list C_j containing a truly corrupted party i .

We define the utility function of detection-averse adversaries as follows.

Definition 8 (Utility of Detection-Averse Adversaries). A detection-averse t -adversary has a utility function U_{DA} that maps an outcome out_A of $G_{\Pi,A}^{\text{eBC-LD}}(n, t)$ to a real value such that

$$U_{\text{DA}}(0, 0, 0) < u_1 \leq U_{\text{DA}}(0, 0, 1) < u_2 < U_{\text{DA}}(1, 1, 1) \quad (2)$$

for any $u_1 \in \{U_{\text{DA}}(0, 1, 0), U_{\text{DA}}(1, 0, 0), U_{\text{DA}}(1, 1, 0)\}$ and $u_2 \in \{U_{\text{DA}}(0, 1, 1), U_{\text{DA}}(1, 0, 1)\}$

Intuitively, a detection-averse adversary gains higher utility when it violates the security requirements of broadcast protocols without having the attack detected.

The security notion is defined based on Definition 5 by replacing utility U with utility U_{DA} of detection-averse adversaries.

Definition 9 (Security of Broadcast against Detection-Averse Adversaries). A broadcast protocol Π is said to be secure against detection-averse t -adversaries if there exists a t -adversary B such that

1. *Security:* The outcome of $G_{\Pi,B}^{\text{eBC-LD}}(n, t)$ satisfies $\text{out}_B = (0, 0, 1)$.
2. *Best response:* For any t -adversary A , the utility satisfies $u_{\text{DA}}(A) \leq u_{\text{DA}}(B)$, where $u_{\text{DA}}(A) = \mathbb{E}[U_{\text{DA}}(\text{out}_A)]$ and out_A is an outcome of the broadcast game $G_{\Pi,A}^{\text{eBC-LD}}(n, t)$.

The best response property of Definition 9 guarantees that no adversary has an attack strategy A that violates either validity/agreement without its attack being locally detected.

Our Protocol

The construction idea for achieving the optimal worst-case round complexity is that we do not perform the main part of Π_{LD} after round 4. The protocol is presented in Fig. 4.

We show that Π_{DA} satisfies validity.

Lemma 5. For any $t < n$, Π_{DA} satisfies validity for t -adversaries.

Proof. To prove validity, suppose for contradiction that sender s with message m is not corrupted and some party $i \in H$ outputs $m' \neq m$ in the execution of Π_{DA} . Since s is honest, the validity of the DS protocol implies that $v_{\text{aux}}^j = m$ for every $j \in H$. Thus, the output of party i should be $m' \neq \perp$. Honest party i outputs such m' by collecting $t + 1$ valid PoAs in round 4. However, since s is honest, no party can generate a valid signature for message m' , a contradiction. $\square$

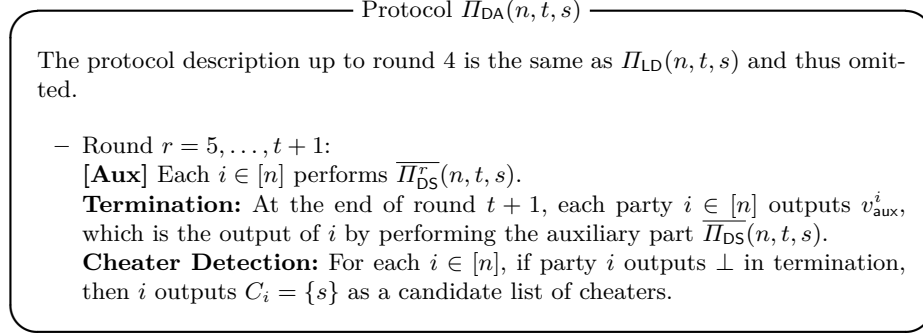


Fig. 4. Our Protocol against Detection-Averse Adversaries

Combining the above lemma with Lemma 1, we have the following corollary.

Corollary 1. *For any $t < n$, Π_{DA} has local detectability against t -adversaries.*

To prove the security against detection-averse adversaries, we show that Π_{DA} satisfies *weak* agreement.

Lemma 6. *For any $t < n$ and t -adversary A , in the execution of $\mathbf{G}_{\Pi_{\text{DA}}, A}^{\text{eBC-LD}}(n, t)$, every output value val_i of party $i \in H$ satisfies $\text{val}_i \in \{m, \perp\}$ for some m .*

Proof. Suppose for contradiction that there exists messages m and $m' \neq m$ and $i, j \in H$ such that $\text{val}_i = m \neq \perp$ and $\text{val}_j = m' \neq \perp$. Note that $\text{val}_i = m$ occurs either in the following two cases:

- (a) Party i outputs m in round 4 by collecting $t + 1$ PoA_m^ℓ for different ℓ .
- (b) Party i outputs m in round $t + 1$ since $v_{\text{aux}}^i = \{m\}$.

Similarly, we consider cases (a') and (b'), which describe the conditions under which j outputs m' .

Lemma 2 excludes cases (a) and (a') since $m \neq m'$.

Now consider cases (a) and (b'). Since there exist $t + 1$ valid PoAs for m , some party $k \in H$ sent PoA_m^k to all parties in round 4. Since PoA_m^k contains $t + 1$ valid PoD_m^ℓ for different ℓ , some party $k' \in H$ must have sent a valid $\text{PoD}_m^{k'}$ to all parties in round 3. This implies that some party $k'' \in H$ received $(m, \sigma_s(m))$ in round 1, and hence k'' performs $\overline{\Pi_{\text{DS}}^2}(n, t, k'')$ in the auxiliary part of round 2. By the validity of the DS protocol, it follows that for every $\ell \in H$, we have $v_{\text{aux}}^\ell \in \{m, \perp\}$. This contradicts the assumption (b') that j outputs m' , since $v_{\text{aux}}^j = \{m'\}$.

Finally, consider cases (b) and (b'). The assumption implies $v_{\text{aux}}^i = \{m\}$ and $v_{\text{aux}}^j = \{m'\}$. However, it contradicts the agreement property of the DS protocol. Therefore, the statement follows. $\square$

We prove the security against detection-averse adversaries.

Theorem 4. *For any $t < n$, Π_{DA} is secure against detection-averse t -adversaries. It has round complexity 4 against detection-averse adversaries and $t + 1$ against worst-case adversaries.*

Proof. We consider the same adversary B as in the proof of Theorem 3. To prove the best response property, we need to show that $u_{\text{DA}}(A) \leq U_{\text{DA}}(0, 0, 1)$ for any t -adversary A . Inequality (2) implies that, to violate the best response, A needs to achieve either $U(0, 1, 1)$, $U(1, 0, 1)$, or $U(1, 1, 1)$. Since Π_{DA} satisfies validity by Lemma 5, A cannot achieve $\text{validity}_{\text{fail}} = 1$. Thus, A needs to achieve the case of $U(0, 1, 1)$.

To achieve $\text{agreement}_{\text{fail}} = 1$, adversary A needs to corrupt the sender s , because, otherwise, every honest party outputs the same value as the sender's input due to the validity of Π_{DA} . To achieve $\text{undetected} = 1$, A needs the situation that

$$\begin{aligned} & \text{agreement}_{\text{fail}} = 1 \text{ and there is no honest party } k \text{ who outputs } C_k \neq \emptyset \\ \Leftrightarrow & \text{agreement}_{\text{fail}} = 1 \text{ and there is no honest party who outputs } \perp \\ \Leftrightarrow & \exists i, j \in H \text{ s.t. } \text{val}_i \neq \text{val}_j \text{ and } \forall k \in H, \text{val}_k \neq \perp. \end{aligned}$$

However, Lemma 6 implies that such $(\text{val}_i, \text{val}_j)$ cannot occur. Thus, we have shown that Π_{DA} satisfies the best response property.

The round complexity analysis immediately follows. $\square$

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